MINGI KANG

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EDUCATION

Bowdoin College, ME *August 2022 - May 2026*
Bachelors of Arts in Computer Science with Honors, Minor in Mathematics
Overall GPA: 3.73 / 4.00

Aquincum Institute of Technology, Budapest, Hungary *August - December 2024*
Study Abroad Semester in Computer Science

PUBLICATIONS, PRESENTATIONS, MEDIA

Publications

[1] IGLU: The Integrated Gaussian Linear Unit Activation Function 2026
Mingi Kang, Zai Yang, Jeová Farias Sales Rocha Neto *ECML PKDD*
European Conference, ECML PKDD 2026, Machine Learning and Knowledge Discovery in Databases. Research Track.
arXiv preprint arXiv:2603.06861

[2] Interpolation between Convolution and Attention via K-Nearest Neighbors 2026
Mingi Kang *Undergraduate Honors Thesis*
Bowdoin College Library Digital Collection Honors Catalogue

[3] Structures and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing 2024 *CogSci*
Abhilasha A. Kumar, Mingi Kang, William G. Kronenberger, Michael N. Jones, David B. Pisoni
Proceedings of the 46th Annual Meeting of the Cognitive Science Society (Vol. 46).
DOI: https://escholarship.org/uc/item/7vn9q9hh

In Preparation/Under Review/Technical Report

[4] FCT: Fully Convolutional Transformers with Linear Attention 2026
Mingi Kang, Jackson Codd, Jeová Farias Sales Rocha Neto *In Preparation*

[5] Attention Via Convolutional Nearest Neighbors 2025
Mingi Kang, Jeová Farias Sales Rocha Neto *In Preparation*
arXiv preprint arXiv:2511.14137

[6] Parallel qMRI Reconstruction from 4x Accelerated Acquisitions 2025
Mingi Kang *Technical Report*
arXiv preprint arXiv:2511.18232

Presentations

[1] Convolutional Nearest Neighbors: Reinterpreting Convolution Through K-Nearest Neighbor Selection October 2025
IEEE MIT Undergraduate Research Technology Conference (MIT URTC) 2025, Cambridge, MA *Poster*

[2] Parallel qMRI Reconstruction from 4x Accelerated Acquisitions July 2025
McKelvey School of Engineering Summer Symposium & Poster Palooza, St. Louis, MO *Oral/Poster*

[3] Structures and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing July 2024 *Poster*
Annual Conference of the Cognitive Science Society (CogSci) 2024, Rotterdam, Netherlands

Media

[1] Mingi Kang '26: Advancing Computers' Ability to See and Understand Our World *November 2025*
Article Link

RESEARCH EXPERIENCE

Senior Honors Thesis in Computer Science August 2025 - May 2026
Honors Candidate (Advisor: Prof Jeova Farias) *Brunswick, ME*

- Developed **ConvNN-Triton**, a custom Triton kernel for the ConvNN neighbor selection and aggregation mechanism, achieving 1.16× faster forward pass, 1.36× faster backward pass, 2.32× lower forward peak memory, and 1.38× lower backward peak memory over baseline PyTorch implementation.
- Architected a hybrid branching layer combining spatial (Conv2d) and feature-based (ConvNN) aggregation, achieving up to 0.9% accuracy improvement on ImageNet-1K with ResNet-50 while reducing GFLOPS by 32% and parameter count by 3% compared to standard convolution.
- Extended ConvNN to an attention variant (ConvNN-Attention) that improves over ViT-Base self-attention by 0.7% on ImageNet-1K with 2% fewer GFLOPS and only 28K additional parameters.
- Designed multi-GPU training on Indiana Jetstream 4×H100 GPUs, implementing Distributed Data Parallel (DDP) and mixed-precision training for large-scale ImageNet-1K experiments across ViT and ResNet architectures.

Independent Deep Learning Research Group August 2025 - May 2026
Undergraduate Research Assistant (Advisor: Prof Jeova Farias) *Brunswick, ME*

- Developed **IGLU** (Integrated Gaussian Linear Unit), a parametric activation function derived as a scale mixture of GELU gates under a half-normal mixing distribution, generalizing both ReLU and GELU with a learnable parameter σ for fine-grained control over activation shape.
- Implemented a modular GPT-2 architecture with plug-and-play activation swapping, training on WikiText-103 to benchmark perplexity: IGLU achieves a 0.11 reduction in test perplexity compared to GELU.
- Demonstrated 1-2% accuracy improvements on imbalanced CIFAR-100 classification with ResNet-20 over ReLU and self-gating baselines (GELU, SiLU).

Computational Imaging Group, Washington University in St. Louis May - August 2025
McKelvey Summer Engineering Fellow (Advisor: Prof Ulugbek S. Kamilov) *St. Louis, MO*

- Designed a self-supervised training pipeline using 2× accelerated scans with k-space masking to simulate 4× undersampling, enabling model training without full-sampled ground truth – validated against real 4× accelerated acquisitions at test time.
- Redesigned a deep unfolding U-Net architecture for parallel qMRI reconstruction, achieving a 4× parameter reduction while maintaining reconstruction quality (37 dB PSNR, 0.923 SSIM) on 4× accelerated scans.
- Developed ACS region-specific and coil-instance normalization techniques for undersampled k-space preprocessing, validated on a patient dataset from WashU Medical Center.

Independent Deep Learning Research Group January - May 2025
Undergraduate Research Assistant (Advisor: Prof Jeova Farias) *Brunswick, ME*

- Developed **Convolutional Nearest Neighbor Attention (ConvNN-Attention)**, an efficient attention mechanism using hard k-NN and convolutional aggregation, reducing computational cost by 18% (GFLOPS) with accuracy improvement (2.4%) on CIFAR-10/100.
- Implemented modular PyTorch layers compatible with standard Transformer and Vision Transformer architectures, enabling drop-in replacement for self-attention.

Christenfeld Summer Research Fellowship January - August 2024
Research Fellow (Advisor: Prof Jeova Farias) *Brunswick, ME*

- Developed **Convolutional Nearest Neighbors (ConvNN)** algorithm integrating norm-based k-NN pixel selection into standard convolutional operations for spatial feature learning.
- Built modular PyTorch implementation (1D/2D) with configurable sampling strategies (random, spatial), pixel-shuffling, and positional encoding.

Lexicon Lab, Bowdoin College December 2022 - May 2024
Undergraduate Research Assistant (Advisor: Prof Abhilasha Kumar) *Brunswick, ME*

- Investigated lexical retrieval processes in prelingually deaf cochlear implant users through computational cognitive modeling, contributing to published CogSci 2024 conference proceedings paper (second author).
- Extended Python package, *Forager* with joint semantic word embeddings (word2vec, speech2vec) for quantitative analysis, revealing differential reliance on speech-derived representations.

TEACHING/MENTORING EXPERIENCE

Bowdoin College Baldwin Center for Learning and Teaching — August 2025 - May 2026
Quantitative Tutor — *Brunswick, ME*

- Provide one-on-one tutoring in computer science (algorithms, data structures, AI), mathematics (linear algebra, probability, statistics), and economics (microeconomics, macroeconomics).
- Support students in debugging code, understanding theoretical concepts, and developing problem solving strategies for quantitative coursework.

Bowdoin College Mathematics Department — August 2025 - May 2026
Learning Assistant (Teaching Assistant) for Statistics and Data Science — *Brunswick, ME*

- Conduct weekly learning assistant hours supporting 20+ students with R programming, statistical analysis, hypothesis testing, data visualization, and data cleaning.
- Guide students through applied projects involving real-world datasets, emphasizing statistical thinking and reproducible analysis.

Bowdoin College Computer Science Department — January 2025 - May 2026
Learning Assistant (Teaching Assistant) for Intro to Computer Science — *Brunswick, ME*

- Lead weekly learning assistant hours for 40+ students covering Python fundamentals and object-oriented programming.
- Mentor students on debugging techniques, code organization, and developer best practices.

Bowdoin College Career Exploration and Development Group — January 2025, January 2026
Sophomore Bootcamp Leader & Pod Leader — *Brunswick, ME*

- Design and deliver workshops on career development for 400+ sophomores, covering resume, writing, behavioral interviews, internship navigation, personal finance, academic research, graduate school, and fellowships.
- Train and manage a team of 7 senior mentors, overseeing their preparation to deliver specialized workshops and career curricula to the sophomore class.
- Mentored 5 student teams in inaugural Sophomore Bootcamp Hackathon, providing technical guidance on full-stack development, API integration, and project management.

RELEVANT COURSEWORK

Computer Science: Deep Learning for Computer Vision, Operating Systems, Artificial Intelligence, Computer Systems, Algorithms, Data Structures, Data Science, Cryptography, Computational Game Theory

Mathematics: Linear Algebra, Multivariable Calculus, Probability, Statistics, Mathematical Reasoning, Advanced Probability and Statistics

TECHNICAL SKILLS

Programming Languages	Python, R, Java, C, C++, SQL, JavaScript, Kotlin
ML & DL	PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, SciPy, OpenCV
Visualization	Matplotlib, Seaborn, ggplot2
Developer Tools	Git, GitHub, Conda, Bash, Overleaf/LaTeX, Markdown
HPC & Parallel Computing	Slurm (sbatch), IBM LSF (bsub), Indiana Jetstream, Distributed Data Parallel (DDP), Mixed-Precision Training (AMP)
GPU Computing	CUDA, Triton kernel development NVIDIA A100, H100, GH200, RTX 5090, RTX 3080, A6000, Pro 6000;
Languages	English (Native), Korean (Native), German (Conversational)

AWARDS AND HONORS

1. Computer Science Senior-Year Prize — *Institutional*
   Bowdoin College — May 2026
   - Awarded to a senior judged by the Department of Computer Science to have achieved the highest distinction in the major program in computer science.

2. Hastings Student AI Award ($ 330) — *Institutional*
   Bowdoin College Hastings Initiative for AI and Humanity (HIAIH) — March 2026
   - Awarded to support advanced GPU computing training through NVIDIA's Deep Learning Institute, with a focus on CUDA/C++ acceleration techniques applied to the Convolutional Nearest Neighbors senior honors project.

3. CRA Outstanding Undergraduate Researcher Award, Honorable Mention — *National*
   Computing Research Association — January 2026
   - Prestigious national honor recognizing exceptional undergraduate computing researchers; first student from Bowdoin College to receive this award. Website Link

4. John L. Roberts Fall Research Award ($ 2,463) — *Institutional*
   Bowdoin College — October 2025
   - Awarded for senior honors project on Convolutional Nearest Neighbors & Convolutional Nearest Neighbors Attention.

5. Best Poster Presentation Award ($ 100) — *Institutional*
   Washington University in St. Louis — July 2025
   - Awarded for best poster presentation among McKelvey Engineering summer research fellows at Washington University in St. Louis.

6. McKelvey Engineering Summer Research Fellowship ($ 7,200) — *Institutional*
   Washington University in St. Louis — May 2025
   - Awarded to conduct 10-week full-time research with the Computational Imaging Group.

7. Allen B. Tucker Computer Science Research Prize ($ 50) — *Institutional*
   Bowdoin College — May 2025
   - Awarded to a computer science student for excellence in summer research.

8. CRA Undergraduate Research Award ($ 4,000) — *Local*
   Last Mile Education — April 2025
   - Awarded to support summer research costs for computational imaging research at Washington University in St. Louis.

9. NYC STEM Award ($ 1,500) — *Local*
   Last Mile Education — March 2025
   - Merit-based grant for technical equipment and research support.

10. Google AI 2024 Award ($ 595) — *Local*
    Last Mile Education — January 2025
    - Awarded for independent semester research extending prior work from the Christenfeld Summer Research Fellowship.

11. Christenfeld Summer Research Fellowship ($ 4,800) — *Institutional*
    Bowdoin College — April 2024
    - Awarded to conduct 8-week full-time research in computer vision and deep learning under faculty mentorship.

12. NSF Student Faculty Research Fellowship ($ 1,800) — *National/Institutional*
    Bowdoin College, National Science Foundation — January 2024
    - Awarded for computational cognitive science research focused on modeling search and retrieval within the mental lexicon.

13. QuestBridge National College Match Finalist — *National*
    QuestBridge — September 2021
    - Recognized as a finalist for the QuestBridge program, which connects high-achieving students from low-income, first-generation backgrounds with full scholarships to top colleges.